

Ollama

Ollama

1. Large Language Models (LLMs)
2. Ollama Installation
 - Script Installation
3. Using Ollama
4. Uninstall Ollama
- References

Demo Environment

Development Board: Raspberry Pi 5

SD (TF) Card: 64GB (16GB or more; the larger the capacity, the more models you can try)

```
Raspberry Pi 5 (16GB RAM): Runs models with 14B parameters or less
Raspberry Pi 5 (8GB RAM): Runs models with 8B parameters or less
Raspberry Pi 5 (4GB RAM): Runs models with 3B parameters or less
Raspberry Pi 5 (2GB RAM): Runs models with 1B parameters or less
Raspberry Pi 5 (1G RAM): Cannot run offline large models. Please refer to the online large model routines.
```

Ollama is an open-source tool designed to simplify the deployment and operation of large language models, allowing users to use high-quality language models in local environments.

1. Large Language Models (LLMs)

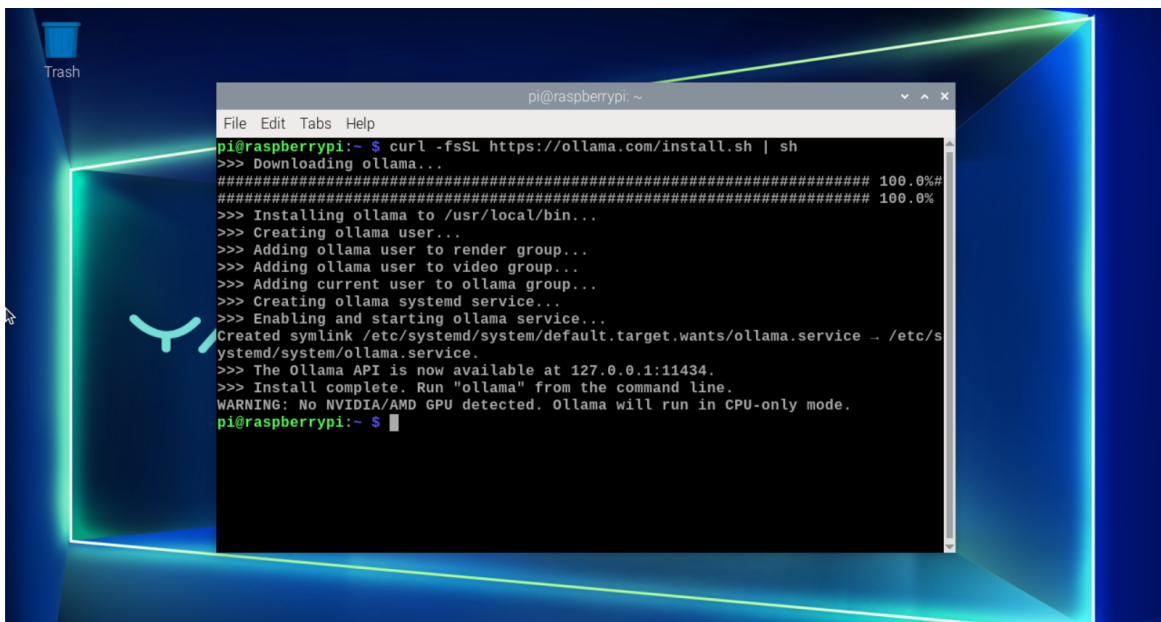
Large language models (LLMs) are a type of advanced text generation system based on artificial intelligence technology. Their main feature is their ability to learn and understand human language using large amounts of training data and generate natural and fluent text.

2. Ollama Installation

Note: Failure to pull the Ollama package is normal. You can try multiple times or use a proxy to pull the package. You will need to search online for proxy installation instructions.

Script Installation

```
sudo apt install curl -y
sudo curl -fsSL https://ollama.com/install.sh | sh
```

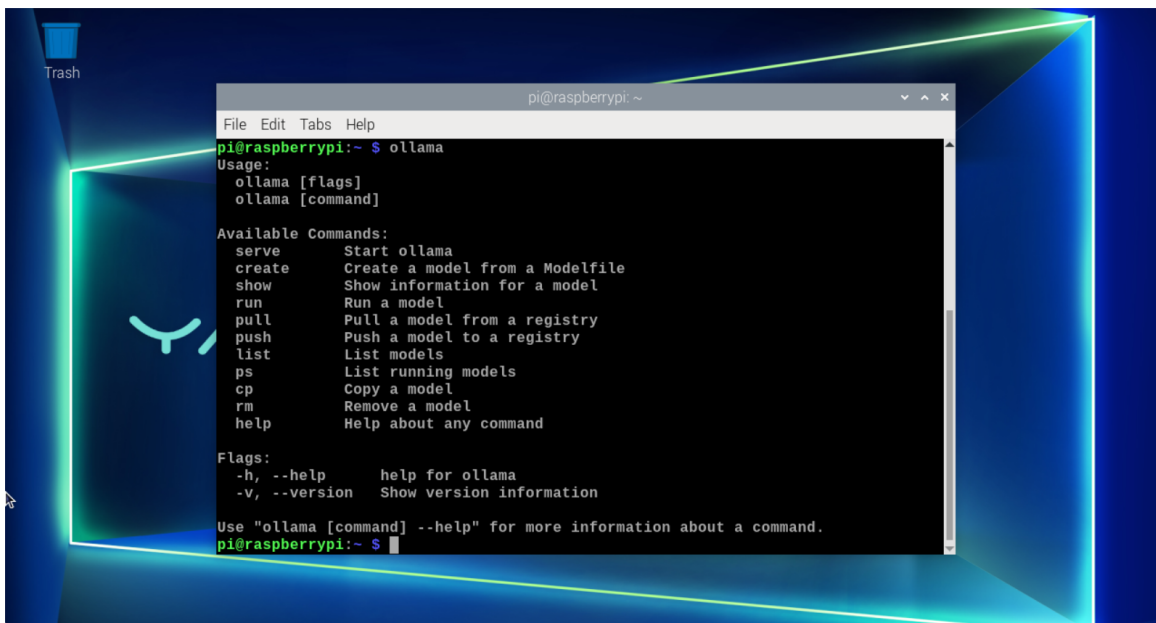


The entire installation process takes a while, so be patient. When the Raspberry Pi terminal displays "Install complete," Ollama has been successfully installed!

The warning indicates that no NVIDIA/AMD GPU was detected, and Ollama will run in CPU mode. We can ignore this prompt.

3. Using Ollama

Type "ollama" in the terminal to see the prompt:



Command	Function
ollama serve	Starts ollama
ollama create	Creates a model from a model file
ollama show	Displays model information
ollama run	Runs a model
ollama pull	Pulls a model from the registry
ollama push	Pushes a model to the registry
ollama list	Lists models
ollama ps	Lists running models
ollama cp	Copys a model
ollama rm	Deletes a model
ollama help	Get help on any command

4. Uninstall Ollama

- Deleting a service

```
sudo systemctl stop ollama
sudo systemctl disable ollama
sudo rm /etc/systemd/system/ollama.service
```

- Deleting files

```
sudo rm $(which ollama)
```

- Deleting model and service users and groups

```
sudo rm -r /usr/share/ollama
sudo userdel ollama
sudo groupdel ollama
```

References

Ollama

Official website: <https://ollama.com/>

GitHub: <https://github.com/ollama/ollama>